

When marginal electricity prices cannot coordinate electric-bus schedules

Complete-fleet certificates and a replenished-fleet counterexample

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Abstract

Electric-bus charging combines continuous energy decisions with indivisible service assignments. A smooth electricity supply cost therefore need not admit marginal prices that support a physically implementable fleet schedule. We formulate the relevant comparison using the convex hull of complete fleet schedules, with the same service, battery, charging and operating-cost assumptions in the planner and price-response models. A classical convexification argument translates the planning gap into a lower bound on every physical schedule's regret at its own marginal price. We give a fully replenished two-service construction with continuous charging, equal net energy purchases and explicit shared power limits. Its exact physical and convexified costs are 97 and 94.8875 synthetic currency units, respectively. At the physical optimum, own-price fleet regret is 13; at the common convex-hull price, fleet lost-opportunity cost is zero and supplier lost-opportunity cost is 2.1125. The example separates physical dispatch, cost at mean load and participant accounting. A complementary synthetic reoptimization diagnostic compares cold optimization, retained columns and an analytic price-shift proposal under fresh certificates. Forty-four of 45 cells meet the frozen acceptance rule; a pricing time-limit failure remains in the record. These results establish an inspectable mechanism and computational baseline, while operational prevalence, market calibration and a funded coordination mechanism remain open.

Keywords: electric-bus scheduling; convex-hull pricing; charging coordination; lost-opportunity cost; column reuse; physical feasibility.

1. Introduction

An electric-bus operator must deliver mandatory passenger services while choosing vehicle duties and charging schedules. Even when the amount of electricity charged in each interval is continuous, moving a service between vehicles changes a discrete duty structure. Electricity prices can encourage a different charging pattern or fleet assignment, but the resulting pattern changes the marginal system cost that motivated those prices. This raises a question distinct from finding a low-cost schedule under an exogenous tariff: can a realizable fleet plan be a best response to the marginal electricity prices associated with its own load?

This paper studies that question through complete fleet schedules. A complete schedule contains all mandatory services, their assignment to vehicles, physically feasible charging and the associated intrinsic operating cost. Its convexification permits averages of these complete plans. Such an average is useful for lower bounds and pricing analysis, but need not be a daily dispatch. In particular, evaluating a convex electricity cost at the average load is not the same as averaging the costs of physical daily loads.

The economic principle is established in nonconvex market optimization. Our contribution is a transparent fleet interpretation, a fully replenished physical counterexample and an evidence structure that separates exact arguments, numerical optimization and operational assumptions. We also evaluate simple schedule reuse before proposing a learned acceleration. The paper does not claim a new general welfare theorem, a new principle of convex-hull pricing, or a demonstrated learning improvement.

The distinction is practically relevant for interpreting optimization output. A small convexified objective does not by itself establish an implementable low-cost plan. A small system-cost gap need not imply a similarly small incentive at the realized marginal price. A payment equal to a fleet's lost opportunity does not by itself fund the supply side or define a budget-balanced institution. Each statement requires a specified physical model, price and deviation right.

2. Relation to existing work

Electric-vehicle scheduling with charging and electricity-related objectives is a developed field. Wu et al. [1] consider multiple depots, time-of-use prices and peak-load reduction, with branch-and-price and trip-chain reuse. De Vos et al. [2] incorporate partial charging and capacitated charging stations. Parmentier et al. [3] develop scalable route and recharge scheduling through column generation, and ten Bosch et al. [4] combine simulated annealing with integer-programming recombination. These works establish important operational and algorithmic precedents. Our central relaxation uses complete fleet plans, which must be distinguished from a master whose columns are individual vehicle duties.

Bus aggregation and electricity-market participation are also established. Zoltowska and Lin [5] aggregate bus charging constraints into auction bids and disaggregate accepted plans. Maldonado and Saumweber [6] examine pricing rules with electric-vehicle participants; their empirical comparison concentrates on IP and extended locational marginal pricing. We do not describe that comparison as an empirical exact-convex-hull benchmark.

Madani et al. [7] connect nonconvex bids, dual prices and aggregate lost-opportunity compensation. Andrianesis et al. [8] compute convex-hull prices using Dantzig-Wolfe decomposition over complete participant schedules. Hümbes et al. [9] emphasize the importance of complete participant descriptions for equilibrium interpretation. These are direct antecedents for our certificate perspective. Bichler et al. [10] distinguish stability, individual rationality and financial balance; we retain that distinction when reporting fleet and supply accounting.

For repeated optimization, Sugishita et al. [11] compare learned warm starts with strong initialization and column-prepopulation baselines. That motivates a conservative computational question: after retaining replay-valid physical columns, how much useful work remains before a fresh certificate is obtained? An analytic tariff shift is another inexpensive conceptual baseline, although solving its proposal oracle still has a computational cost. The full source matrix records inspected versions and access limits; this focused review is not an exhaustive novelty census.

3. Physical and economic model

Let S be the same nonempty compact set of complete physically feasible schedules in every model. A schedule s has intrinsic operating cost $c(s)$ and charging load $L(s)$, measured in kWh per modeled interval. The load vector and cost are continuous on each discrete duty structure; finitely many structures with bounded charging give a sufficient compactness condition. Mandatory service cannot be discarded to improve the objective.

Let F be a supply-cost function defined on the nonnegative supply domain. Assume that its restriction to that domain is proper, closed and convex, and that it admits a differentiable convex extension to an open neighborhood of the feasible fleet-load hull. These global supply-domain assumptions support the conjugate accounting below; the price-support inequality itself uses only convexity and differentiability near the feasible hull. Define $h(s)=c(s)+F(L(s))$. The physical planner value is $D=\min h(s)$. The complete-schedule convexified value CH minimizes $c+F(L)$ over the convex hull of all physical pairs $(c(s),L(s))$. The planning gap is $\Delta=D-CH$, which is nonnegative. This hull retains within-fleet resource constraints in every component before mixing.

At a posted price p , the fleet response value is $V(p)=\min \text{ over } S \text{ of } c(s)+p \cdot L(s)$. At a physical schedule's own price $p_s=\text{grad } F(L(s))$, define $r(s)=c(s)+p_s \cdot L(s)-V(p_s)$. This is price-taking regret: the alternative schedule is evaluated while the posted price stays fixed. It is not a strategic bill comparison in which the operator anticipates the change in price caused by its alternative load.

The single-fleet interpretation is deliberate. If independent operators share chargers, a unilateral deviation must specify whether residual physical capacity is reserved, whether capacity rights are traded, or whether congestion is priced. The convex hull of jointly feasible whole-fleet plans cannot silently substitute for each independent operator's unrestricted opportunity set.

4. Price-support and accounting diagnostics

The following fleet specialization uses the classical convexification and duality perspective of [7-9]; it is not a new general pricing theorem.

Proposition 1. Under the stated common-model, compactness and convexity assumptions, every physical schedule satisfies $r(s) \geq h(s)-CH \geq D-CH$. Moreover, $D=CH$ if and only if a physical schedule is a best response at its own marginal price. When $D=CH$, every physical planner optimizer has that property.

$$r(s) \geq h(s) - CH \geq D - CH = \Delta.$$

Proof. A linear functional has the same minimum over the physical cost-load set and its convex hull. For any hull point $(c_{\text{bar}}, L_{\text{bar}})$, its linear private objective is at least $V(p_s)$. The supporting-plane inequality gives $F(L_{\text{bar}}) \geq F(L(s)) + p_s \cdot (L_{\text{bar}} - L(s))$. Adding these inequalities and minimizing over the hull yields $CH \geq h(s) - r(s)$. If $r(s)=0$, the chain $CH \geq h(s) \geq D \geq CH$ forces equality. Conversely, a physical planner optimizer is also a hull optimizer when the values agree. The directional derivative of the convex objective toward every physical point is then nonnegative, which is exactly the global best-response inequality at its marginal price.

Attainment is essential to the converse. Equality of infima alone need not produce a physical optimizer. Likewise, support of a best-response correspondence does not prove convergence of an iterative rule that selects one optimizer at every price, or uniqueness of that optimizer.

For numerical work, let $[L_D, U_D]$ and $[L_{CH}, U_{CH}]$ be valid objective enclosures. The planning gap lies in $[L_D - U_{CH}, U_D - L_{CH}]$.

$$\Delta \in [L_D - U_{CH}, U_D - L_{CH}].$$

A positive lower endpoint is the relevant nonexistence diagnostic. A restricted pool's replay-feasible mixture can provide U_{CH} , but its objective is generally not a lower bound for the complete hull. An interval crossing zero leaves the sign unresolved. Numerical enclosures conditional on solver and replay tolerances are not formal exact-arithmetic certificates.

For participant accounting, the supplier may choose any nonnegative load vector, independently of the fleet charging-resource limits. At a common price p for which the conjugate is finite, define the nonnegative-load supply conjugate $F^*(p) = \sup_{L \geq 0} p \cdot L - F(L)$. Fleet LOC is $c(s) + p \cdot L(s) - V(p)$, and supply LOC is $F(L(s)) - p \cdot L(s) + F^*(p)$. Their sum is $h(s) - [V(p) - F^*(p)]$. At a physical planner optimizer and a dual-optimal hull price, under strong duality, this sum equals Δ . At the schedule's own gradient price, supply LOC is zero and fleet LOC equals $r(s)$. These statements condition on different prices. Neither identity proves budget balance, truthfulness, voluntary participation or convergence.

$$\text{LOC}_{\text{fleet}} + \text{LOC}_{\text{supply}} = h(s) - [V(p) - F^*(p)].$$

5. A fully replenished continuous-charging construction

Two mandatory depot-to-depot services consume 15 kWh each. Service A occurs in hour 0-1 and service B in hour 2-3. At most two identical used buses are available. Each starts and ends with 20 kWh, which is also its battery capacity; the reserve is zero. Charging is continuous and lossless. The early window, hour 1-2, permits 10 kWh in total and 10 kW per bus. The native terminal window, hour 3-4, permits 30 kWh in total and 30 kW per bus. There are enough connectors. No other charging, deadheads, auxiliary load or battery degradation is modeled.

These are explicit synthetic assumptions. The terminal window is part of the physical model, rather than a zero-energy passenger service. An unused bus contributes neither energy nor cost. Every used bus must cover a service. With two unsplittable services there are exactly two unlabeled partitions: one bus serves both, or two buses serve one each.

For one bus, early charge x must satisfy $x \geq 10$ to make service B feasible and $x \leq 10$ from shared power, so $x = 10$. It charges 20 kWh in the terminal window. For two buses, the bus serving A can charge any x in $[0, 10]$ early and $15 - x$ at the end; the bus serving B charges 15 at the end. Every bus finishes at its starting SOC. Both structures therefore purchase exactly 30 kWh. The extra bus cannot donate net initial energy.

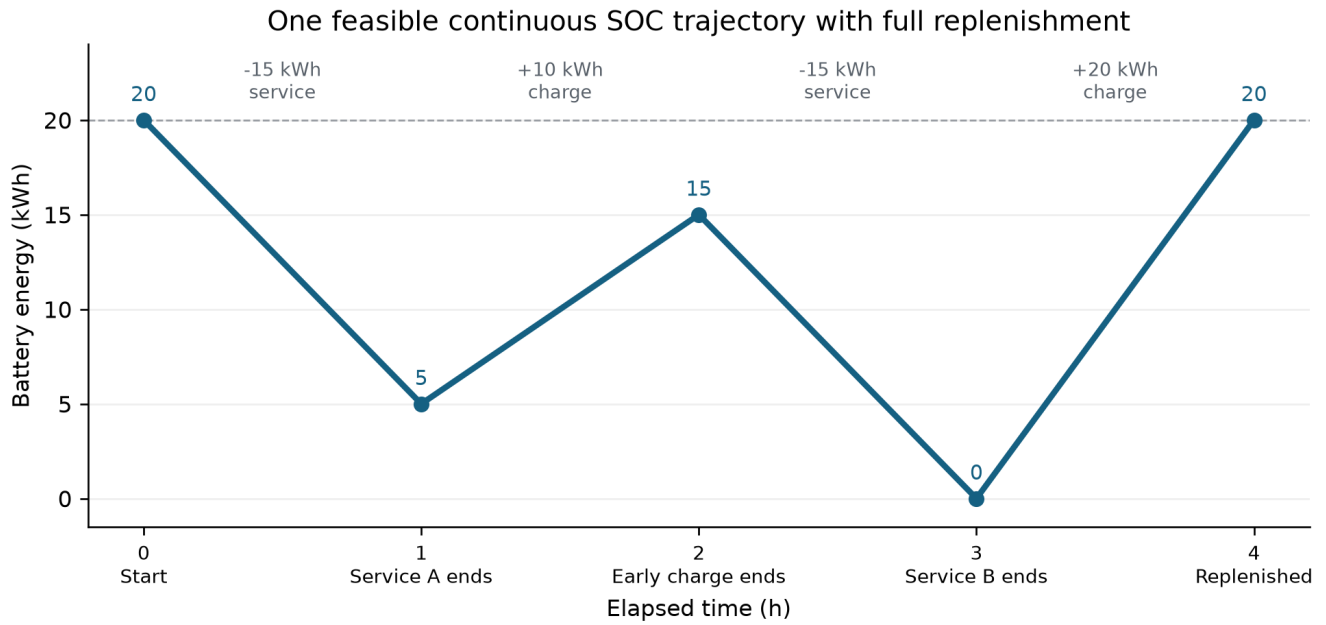


Figure 1. One feasible continuously charged one-bus trajectory. The two services each consume 15 kWh; early and terminal charging add 10 and 20 kWh. The line uses constant consumption and charging rates within each one-hour period, consistent with the stated energy model. Initial and final battery energy are equal. This is an explanatory construction, not a measured bus trace.

For this construction, define the global supply function $F(e,l)=a*e+0.1*(e^2+l^2)$ for all nonnegative early and terminal supply quantities e,l . With total fleet recharge 30, its restriction is $G(x)=F(x,30-x)$. For used-bus cost $f=7$ and early intercept $a=4$, the one-bus cost is 97. The two-bus branch has cost $14+G(x)$, minimized at $x=5$ with value 99. The physical optimum is $D=97$.

$$D = 97, \quad CH = 94.8875, \quad \Delta = \frac{169}{80} = 2.1125.$$

Let λ be the weight on the one-bus structure. The complete projected hull has $0 \leq \lambda \leq 1$, $10\lambda \leq x \leq 10$ and intrinsic cost $14-7\lambda$. These conditions are necessary by convex combination. For $\lambda < 1$ they are sufficient by assigning the remaining two-bus component early charge $(x-10\lambda)/(1-\lambda)$. At $\lambda=1$, the constraints force $x=10$, which is the physical one-bus point. At fixed x the cheapest mixture uses $\lambda=x/10$. The convexified objective is therefore $104-2.7x+0.2x^2$, or $7591/80+(x-27/4)^2/5$.

$$CH = \min_{0 \leq x \leq 10} \left[\frac{7591}{80} + \frac{(x-27/4)^2}{5} \right] = \frac{7591}{80}.$$

Its minimum occurs at $x=6.75$ and $\lambda=0.675$, giving $CH=7591/80=94.8875$ and $\Delta=169/80=2.1125$.

A supporting mixture puts weight $27/40$ on the one-bus schedule with load $(10,20)$ and weight $13/40$ on the two-bus schedule with load $(0,30)$. Both components individually obey the shared limits and full replenishment. The mean load $(6.75,23.25)$ is not assigned a fractional physical fleet in the interpretation. Randomizing those components on different days costs 99.275 on average, rather than 94.8875; Jensen's inequality explains the difference.

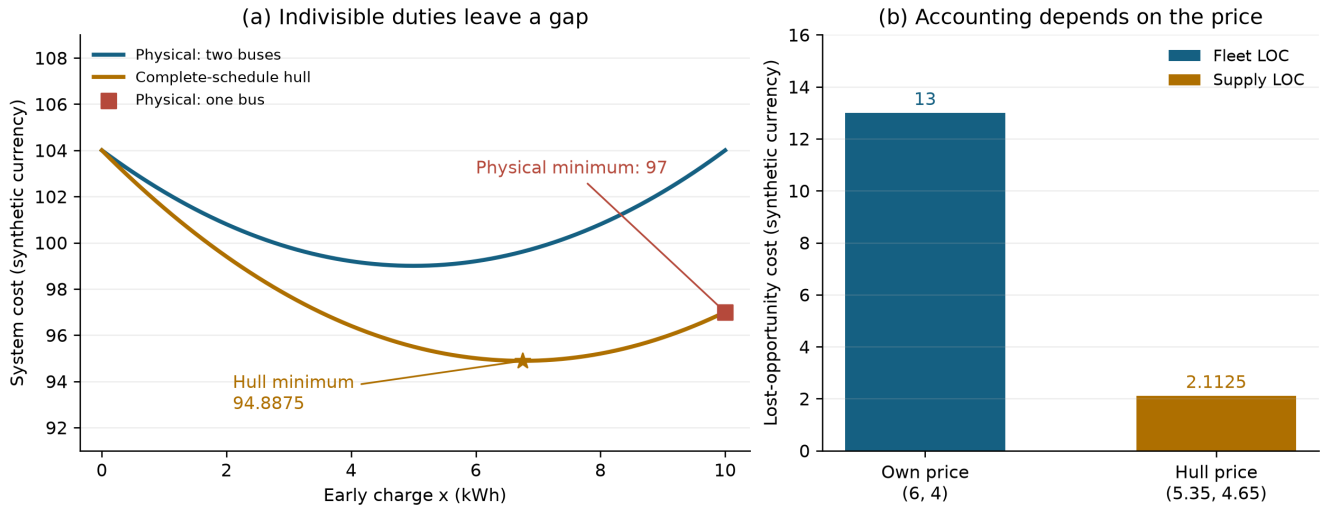


Figure 2. (a) Physical branches and the lower-cost boundary of the complete-schedule hull. (b) Fleet and supplier LOC for the same physical planner optimum, evaluated at two different prices. Prices are synthetic currency/kWh and costs are synthetic currency. The physical point and hull minimum are exact rational results. A hull optimizer is a pricing/lower-bound object, not a realizable fractional bus plan.

At the physical optimum, marginal prices are (6,4). Its private objective is 147; two buses charging only in the terminal window have private objective 134. Thus own-price regret is 13. At the hull price (5.35,4.65), both supporting physical components have private objective 153.5. The physical planner's fleet LOC is then zero. Supply LOC is 2.1125, accounting for the whole planning gap. The supply-side calculation uses $F^*(p)=58.6125$, so the common-price dual value is $153.5-58.6125=94.8875$.

6. Exact parameter checks and evidence quality

The protocol fixed 24 constructed cases before execution: f in $\{1,3,7,10,15,20\}$ and early intercept a in $\{0,2,4,6\}$, holding physics and quadratic coefficient fixed. For each case, the two-bus branch minimizes $G(x)=a*x+[x^2+(30-x)^2]/10$ on $[0,10]$; the hull minimizes $2f-f*x/10+G(x)$. A Fraction-arithmetic driver records both physical schedules and the hull mixture.

An independent implementation reconstructs the same optimization using three physical vertices and quadratic interpolation on the hull edges. It imports no author code and invokes no solver. It checks the full parameter grid, physical coverage, every saved event SOC, individual/shared power limits, exact terminal energy and mixture accounting. All ten deliberate corruptions are rejected. The 24-case grid has 11 positive gaps and 13 exact zeros; these counts describe a fixed analytical design, not a population frequency or a statistical sample.

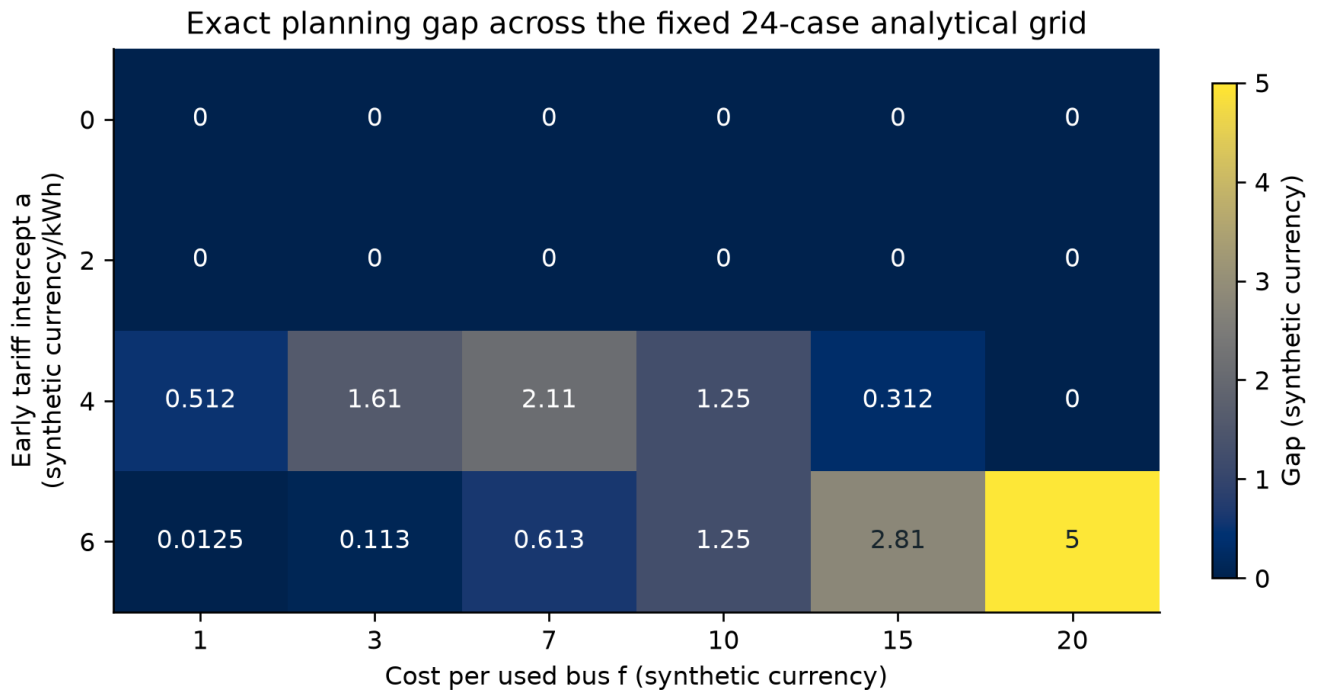


Figure 3. Gaps on the prospectively fixed analytical grid. Labels are rounded; exact rational values are archived. Fleet cost and early supply intercept vary; service demand, charging windows, shared power and boundary energy stay fixed. Values are synthetic currency. The categorical grid is for mechanism inspection and includes all declared cases, including zeros.

The construction establishes existence under fair boundary-energy accounting. Its operational magnitude remains unknown. Reserve requirements, charging losses, finite plugs, time-dependent travel and nonlinear tapering need qualified extensions before making case-study claims. The exact reference model is separate from the existing production EVSP adapter; no full adapter equivalence is inferred from the example.

7. Reoptimization before learning

Repeated certified hull and pricing calculations motivate a computational diagnostic: whether previously feasible complete schedules remain useful when only the tariff intercept changes. This is a baseline study, not a second claimed methodological breakthrough. It compares cold initialization, retention of the same arm's previous physical columns, and retention plus one proposal priced at the previous clean price plus the known intercept change. All arms rebuild tangents, duals and objective bounds. Only fresh clean pricing contributes to the lower certificate; proposal calls count even when they produce duplicates.

The earlier two-service qualification contained four tariff states and three arms. Total pricing calls were 12, 6 and 9 for cold, retained and retained-plus-shift. After initialization, the retained arm needed only the mandatory clean verification. This motivated a harder fixed design rather than immediate learner training.

The new design contains three hand-authored three-service fixtures, five tariff states and three arms, for 45 cells. Two fixtures allow one-/two-bus competition and battery depletion, with different vehicle costs. A third fixes two buses using explicit simultaneous terminal markers and restores both batteries; it is an inventory control, not a native overnight adapter or a variable-fleet cyclic example. Each trajectory contains an unchanged state, early-cheap and

late-cheap changes, and a return to the initial tariff. All cases use continuous charging and multiple opportunities. They do not include shared-power or plug-count constraints.

The acceptance contract uses a fresh clean restricted master, a complete pricing bound, and a replayed true quadratic upper bound, with objective width at most 0.01. Each pricing solve has a ten-second cap and must return OPTIMAL. The total admission budget is 900 seconds, with one solver thread. After the arms finish, a separate complete-structure formulation supplies 15 reference intervals; those solutions never seed the comparison. The reference shares enumeration utilities and the CBC backend, so it is a formulation cross-check, not independent solver software.

All 45 cells were attempted in the first frozen execution. Forty-four certified cells passed their reference overlap checks, and all 15 references completed. The cold return-to-base cell in the lower vehicle-cost depleted fixture returned FEASIBLE at the pricing time cap and remains failed under the protocol. A narrow pricing bound is not substituted retrospectively for its OPTIMAL gate.

An independent implementation, importing no author code or optimization software, reconstructs all 223 pricing attempts with exact rational inventory-flow dynamic programming. Integral flow supplies and capacities make this exact for these continuous linear pricing problems. Exact Fenchel lower bounds computed from the completed clean prices independently support all 44 successful cells, with objective widths below 0.000913. Physical witnesses are checked at a 0.0000001 kWh tolerance. All 19 corruption controls are rejected. The reference witnesses and complete structure enumeration pass, although their tighter native lower bounds remain conditional solver evidence. This audit is specific to the frozen fixtures, not a certification of the general production adapter.

Fixture	Cold pricing calls by state	Retained calls	Retained + shift calls
Depleted, f=20	7, 7, 5, 6, 7 failed	7, 1, 2, 3, 1	7, 2, 3, 4, 2
Depleted, f=26	6, 6, 4, 6, 6	6, 1, 1, 2, 1	6, 2, 2, 3, 2
Replenished, fixed two buses	14, 14, 11, 10, 14	14, 1, 1, 2, 1	14, 2, 2, 3, 2

Table 1. Pricing calls include initial seeds, proposals and clean verification. The failed cell includes attempted work and is not a certified time-to-solution observation. States are base, unchanged, early-cheap, late-cheap and return-base, in that order. Dependent states and hand-built fixtures are not independent population samples.

Across all attempts, the arms use 123, 44 and 56 pricing calls and 99.593, 28.062 and 31.881 complete cell seconds, respectively. At the 14 fixture-state keys where all three arms succeed, the corresponding totals are 116, 43 and 54 calls and 78.506, 27.656 and 31.375 seconds. The cold failure contributes seven attempts and 21.088 seconds to the first view. These are single-run descriptive totals, with success conditioning explicit; they do not estimate a general runtime speedup.

Some retained transitions require two or three clean calls, establishing remaining work on this diagnostic. The analytic proposal did not reduce the subsequent clean-call count in these trajectories. Across twelve warm transitions, retention uses 17 pricing calls. An architecture that always adds one full-pricing proposal and still requires one clean pricing verification has a lower limit of 24 calls, even if every proposal is useful. It therefore cannot improve this

pricing-call metric on the observed fixed trajectory. Direct learned schedule proposals or selective triggering are different architectures; neither is tested here. The remaining discovery work is not evidence that a learner can remove it. A learning experiment would need a separate frozen development/evaluation design, equal final certificates and full offline/online accounting.

8. Operational interpretation and remaining evidence

The intended transportation contribution needs a timetable-based study in addition to exact constructions. That study must identify mandatory trips separately from a provider's solved vehicle blocks; preserve source-row lineage; specify directed and time-dependent deadheads; and state vehicle energy, reserve, terminal recharge and charger-resource assumptions. Historical block membership can provide a feasible reference or warm start, but cannot establish a globally optimal schedule or reveal the provider's objective weights.

A local GIRO audit has identified a small source-matched subproblem with complete service-energy fields, while full-day weekday selection and deadhead translation need explicit treatment. Private source files remain outside the public artifact. Public benchmark sources are being qualified in parallel. No result in this draft is described as an operational benefit or a measurement from either source.

A timetable alone does not identify the supply curvature. Exogenous time-of-use tariffs, a convex incremental supply cost, contractual demand charges and charger scarcity rents are different economic objects. The first operational sensitivity study should report energy, bus count, physical cost bounds, hull bounds, regret bounds and the stated curvature scale separately. Until calibration is defensible, monetary magnitudes must be labeled stylized.

9. Discussion and limitations

The central construction shows that continuous charging does not erase duty indivisibility, even when all net service energy is purchased and every used battery is replenished. It does not imply that large real fleets have substantial unsupported incentives. Fleet aggregation, the relative scale of indivisible duties, supply curvature and access to shared resources can change the effect. Those factors require controlled extensions rather than a claim that the 24 constructed cases represent bus systems generally.

The certificate interpretation is strongest when physical models match exactly. An easier duty-level relaxation, a partially enumerated schedule menu or an aggregate-only capacity constraint can answer a different question. For large cases, it may still be possible to certify a positive gap without enumerating the entire hull: a valid physical lower bound and a replay-feasible mixture upper bound suffice. Failure to prove a positive gap is not proof of equilibrium existence.

The numerical reuse study has a different purpose. It is a development diagnostic, with one execution, small fixtures and strict native-solver termination. It does not estimate a runtime distribution or demonstrate learning effectiveness. The failed cold cell illustrates why full work and failures belong in any speed comparison. Future revisions should add an independent solver check and a prospectively chosen larger design only when they answer a concrete scientific uncertainty.

Finally, price support is an incentive statement under specified rights, not an implementation mechanism. A convex-hull price may support fleet behavior while leaving supplier LOC.

Charging-resource scarcity may require a separate allocation or rent account. Neither can be resolved by relabeling fleet regret as a universally sufficient subsidy.

10. Conclusion

Complete fleet schedules provide the appropriate object for distinguishing physically implementable bus operations from convexified charging plans. A fully replenished continuous-charging example has an exact positive planning gap and sharply different lost-opportunity accounts at own and common hull prices. Reoptimization experiments show that retained schedules are a necessary baseline and that additional proposal work must justify its cost under a fresh certificate. The next empirical task is to determine the magnitude and relevance of these effects on a provenance-qualified timetable while retaining explicit energy and resource assumptions.

Reproducibility and draft status

The exact source and protocol were frozen at Git commit 7bf913a before execution. The harder reuse source, protocol and adversarial tests were frozen at 7d3d764 before execution. Both result trees retain source identities and complete declared cases. The exact result and all 44 successful reuse certificates passed independent reconstruction; the reuse result retains its one failure. The audit found 145 mutable master-log snapshots. The solved tangent sets are recoverable from the frozen control flow; a prospective deep-copy repair at 78bb4c0 preserves future snapshots without replacing any first-run artifact. The independent Fenchel certificates do not depend on that log repair. Figure scripts read the archived result JSON. The source-verified related-work matrix and BibTeX accompany this draft. This version is a substantive working manuscript, not a submission-ready claim of operational validation.

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